

Kosaraju's algorithm

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In computer science, **Kosaraju's algorithm** (also known as the **Kosaraju–Sharir algorithm**) is a linear time algorithm to find the strongly connected components of a directed graph. Aho, Hopcroft and Ullman credit it to an unpublished paper from 1978 by S. Rao Kosaraju. The same algorithm was independently discovered by Micha Sharir and published by him in 1981. It makes use of the fact that the transpose graph (the same graph with the direction of every edge reversed) has exactly the same strongly connected components as the original graph.

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The algorithm

Kosaraju's algorithm works as follows:

- Let G be a directed graph and S be an empty stack.
- While S does not contain all vertices:
 - Choose an arbitrary vertex v not in S . Perform a depth-first search starting at v . Each time that depth-first search finishes expanding a vertex u , push u onto S .
- Reverse the directions of all arcs to obtain the transpose graph.
- While S is nonempty:
 - Pop the top vertex v from S . Perform a depth-first search starting at v in the transpose graph. The set of visited vertices will give the strongly connected component containing v ; record this and remove all these vertices from the graph G and the stack S . Equivalently, breadth-first search (BFS) can be used instead of depth-first search.

Complexity

Provided the graph is described using an adjacency list, Kosaraju's algorithm performs two complete traversals of the graph and so runs in $\Theta(V+E)$ (linear) time, which is asymptotically optimal because there is a matching lower bound (any algorithm must examine all vertices and edges). It is the conceptually simplest efficient algorithm, but is not as efficient in practice as Tarjan's strongly connected components algorithm and the path-based strong component algorithm, which perform only one traversal of the graph.

If the graph is represented as an adjacency matrix, the algorithm requires $O(V^2)$ time.

References

- Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman. *Data Structures and Algorithms*. Addison-Wesley, 1983.

- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. *Introduction to Algorithms*, 3rd edition. The MIT Press, 2009. ISBN 0-262-03384-4.
- Micha Sharir. A strong connectivity algorithm and its applications to data flow analysis. *Computers and Mathematics with Applications* 7(1):67–72, 1981.

External links

- A description and proof of Kosaraju's Algorithm (<http://lcm.csa.iisc.ernet.in/dsa/node171.html>)
- Good Math, Bad Math: Computing Strongly Connected Components (<http://scienceblogs.com/goodmath/2007/10/30/computing-strongly-connected-c/>)

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